Multilayer Neural Networks

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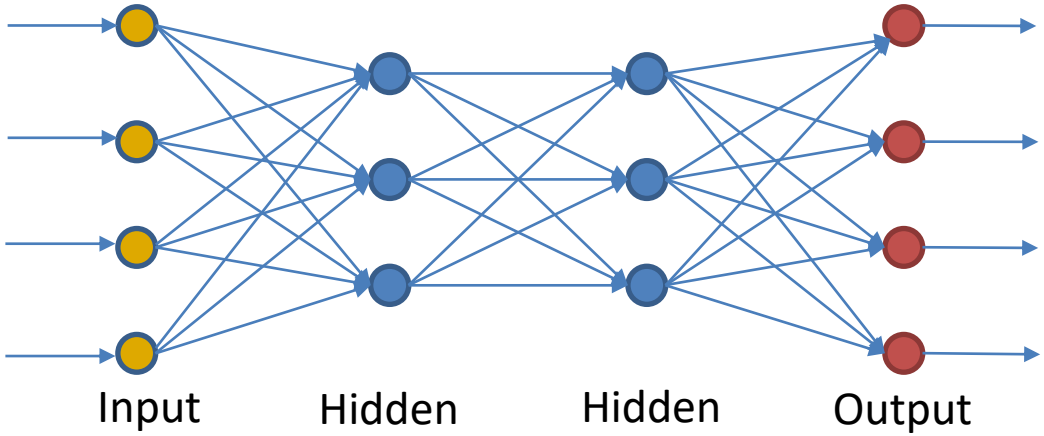
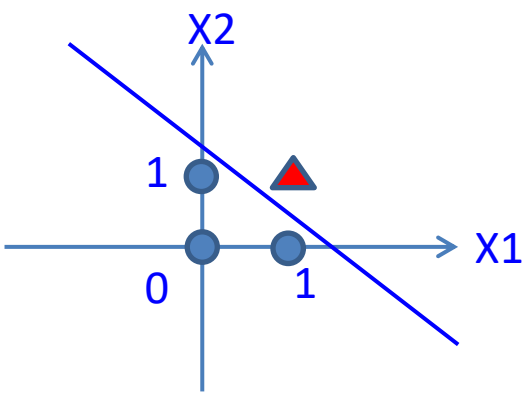
Multilayer neural networks

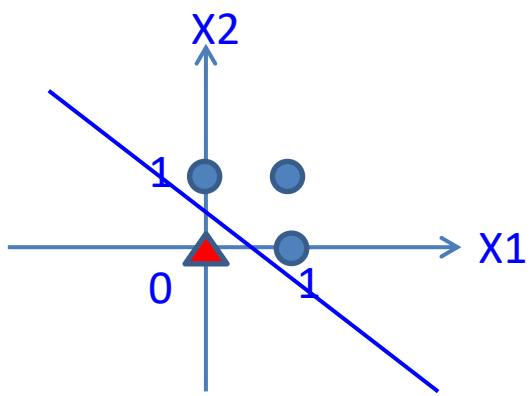


- **Input layer** accepts input signals from the outside world and redistributes these signals to all neurons in the hidden layer.
- **Hidden layers** detect the features of the input patterns by adjusting weights of the neurons.
- **Output layer** accepts signals from the hidden layer and establishes the output pattern of the network.
- Multilayer neural networks can solve the **non-linearly separable problem**.

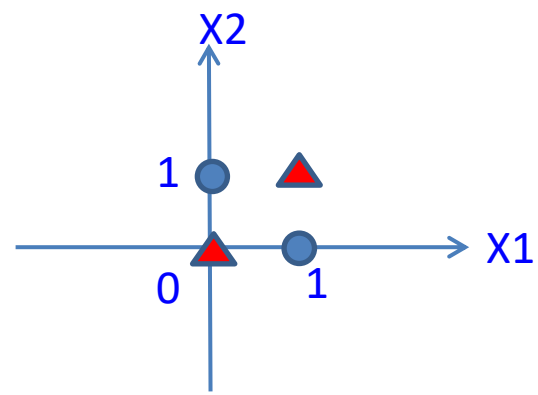
Nonlinearly separable problem



X1 AND X2



X1 OR X2



X1 XOR X2

Linearly separable

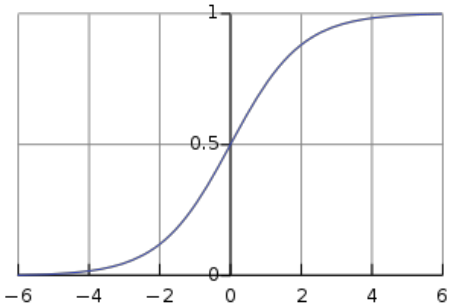
Non-linearly separable

- With one hidden layer, we can represent the continuous and simple discontinuous functions.
- With two hidden layers, even discontinuous function can be represented.
- Most practical application use three-layer neural network (1-1-1: input-hidden-output) to learn patterns.
- The most popular learning algorithm is "backpropagation" (Bryson & Ho, 1969).
- Each neuron determines its output Y as the following:

Threshold (Interception)

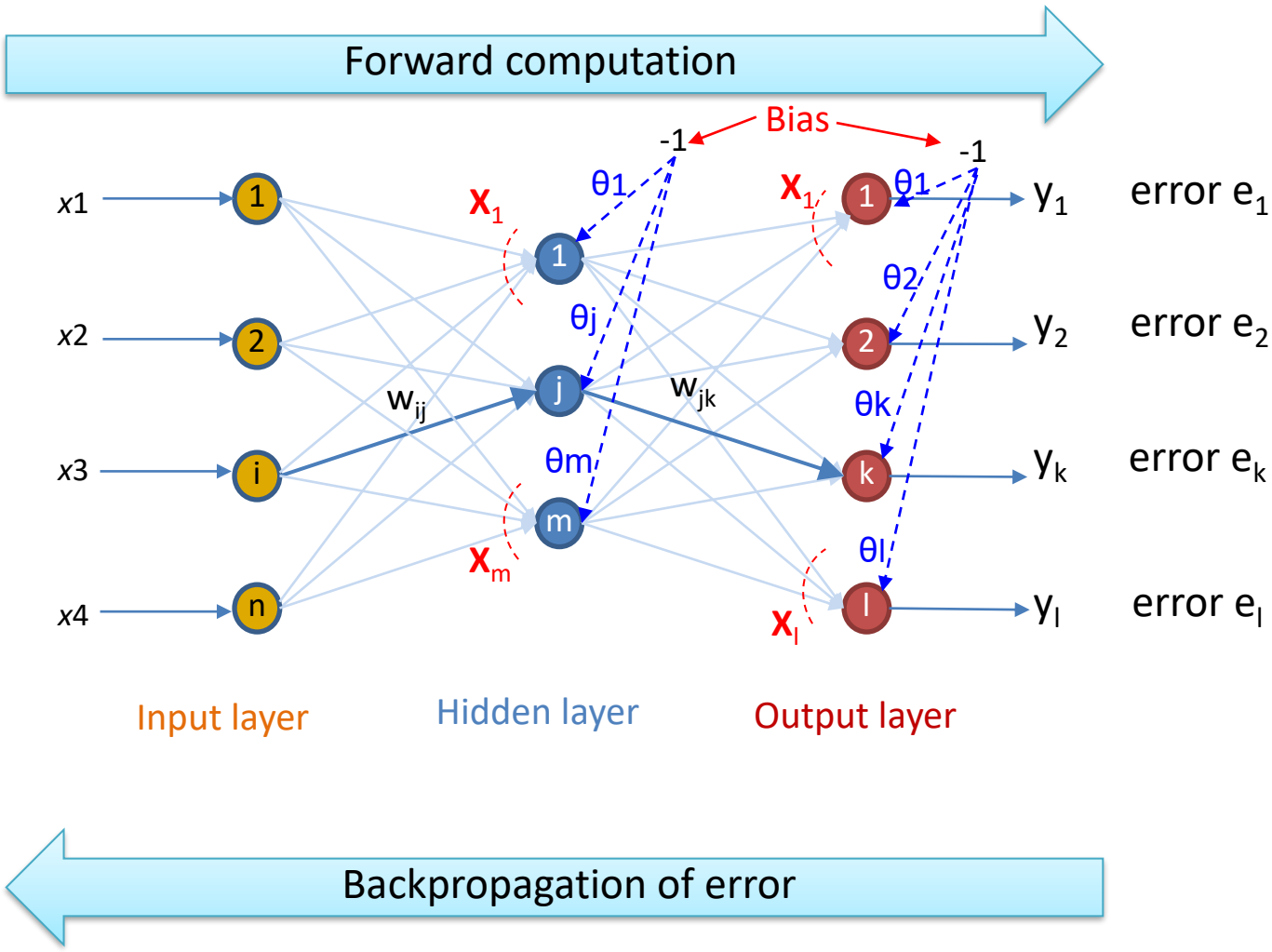
$$X = \sum_{i=1}^{n} x_i w_i - \theta,$$

$$y = \frac{1}{1+e^{-x}} \quad ; 0 < y < 1 \qquad \text{(Sigmoid)}$$



Notations

$$X = \sum_{i=1}^{n} x_i w_i - \theta,$$



Backpropagation for multilayer NN

- To propagate error signals, we start at the output layer and work backward to the hidden layer.

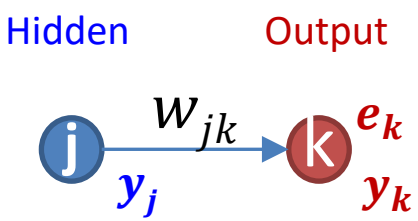
- The error signal at the output of neuron k at iteration p is defined by:

$$e_k(p) = yd_k(p) - y_k(p)$$

Where $yd_k(p)$ is the desired output of neuron k at iteration p.

- To update weights, the weight correction (Δw) is adjusted to the previous weights.

$$w_{jk}(p+1) = w_{jk}(p) - \Delta w_{jk}(p),$$

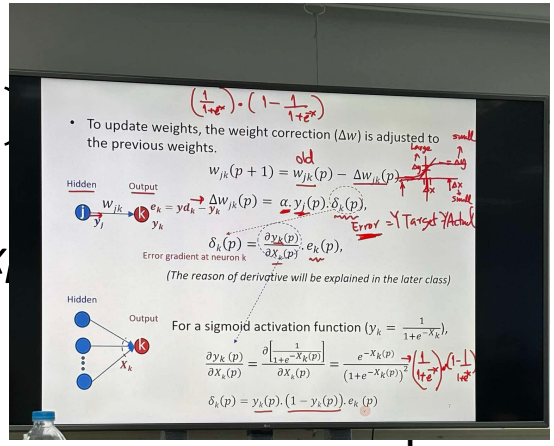


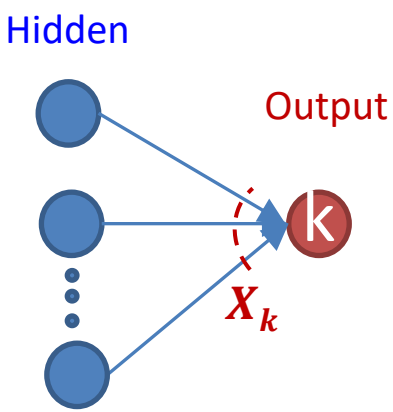
$$\Delta w_{jk}(p) = \alpha . y_j(p) . \delta_k(p),$$

$$\delta_k(p) = \frac{\partial y_k(p)}{\partial X_k(p)} . e_k(p)$$

Error gradient at neuron k

(The reason of derivative will be ex



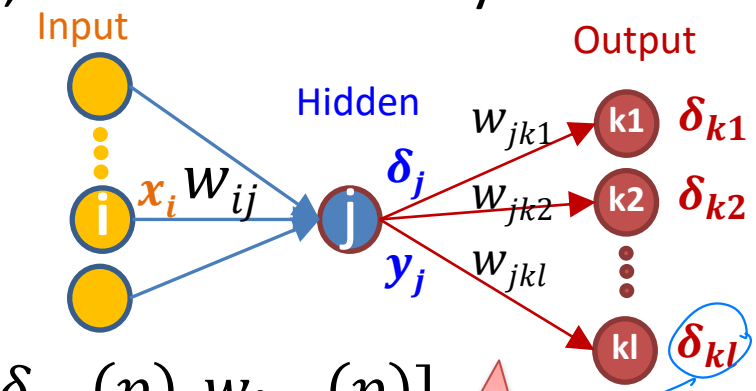


For a sigmoid activation function ($y_k = \frac{1}{1+e^{-X_k}}$),

$$\frac{\partial y_k(p)}{\partial X_k(p)} = \frac{\partial\left[\frac{1}{1+e^{-X_k(p)}}\right]}{\partial X_k(p)} = \frac{e^{-X_k(p)}}{\left(1+e^{-X_k(p)}\right)^2}$$

$$\delta_k(p) = y_k(p).\left(1 - y_k(p)\right).e_k(p)$$

- To update weights in the hidden layer, we do similar way:



$$w_{ij}(p+1) = w_{ij}(p) - \Delta w_{ij}(p),$$

$$\Delta w_{ij}(p) = \alpha . x_i(p) . \delta_j(p),$$

$$\delta_j(p) = \underbrace{y_j(p).\left(1 - y_j(p)\right)}_{\frac{\partial y_j(p)}{\partial X_j(p)}} . \underbrace{\sum_{q=1}^{l}[\delta_{kq}(p).w_{jkq}(p)]}$$

เอา gradiant ที่หามาของแต่ละ node คูณ weight เข้ามา

gradiant

$$\delta_k(p) = y_k(p).(1 - y_k(p)).e_k(p)$$

แล้ว sum ทั้งหมด เป็นตัวเดียว

แล้วจะได้ δ_j (เลขต่ำ j) ก็คือ gradiant node j

Where l is the number of neurons in the output layer.

$$y_j(p) = \frac{1}{1+e^{-X_j(p)}},$$

$$X_j(p) = \sum_{i=1}^{n}[x_i(p).w_{ij}(p)] - \theta_j$$

This all is back popagation

Complete training algorithm

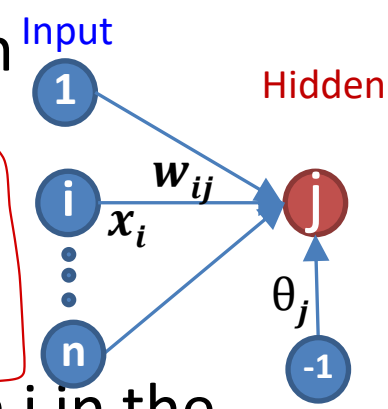
Step1: Weights Initialization

- Set all weights and thresholds (θ) to random number uniformly distributed inside a small range, e.g., (-0.5, +0.5).

good practice ควรมีค่าบวกและลบ และอยู่ในช่วงแคบๆ

Step2: Activation (Feed forward computation)

- Calculate the actual outputs of neurons in the hidden layer.



$$y_j(p) = sigmoid\left[\sum_{i=1}^{n}\left(x_{i(p)}.w_{ij(p)}\right) - \theta_j\right]$$

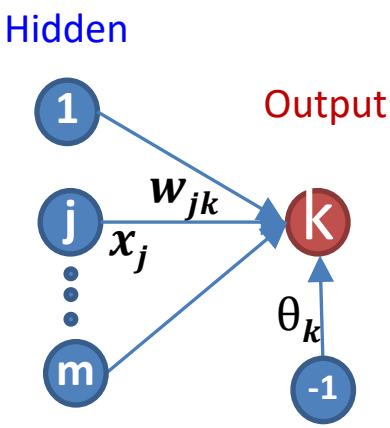
Where n is the number of inputs of neuron j in the hidden layer.

คูณ input แล้วคูณ weight → summation → แล้วเข้า sigmoid

- Calculate the actual outputs of neurons in the output layer.

$$y_k(p) = sigmoid\left[\sum_{j=1}^{m}\left(x_{j(p)}.w_{jk(p)}\right) - \theta_k\right]$$

Where m is the number of inputs of neuron k in the output layer.



Step3: Weight training (Backpropagation)

- Calculate the error gradient of neurons in the hidden-output layer.

ค่า target ค่า actual

$$e_k(p) = yd_k(p) - y_k(p),$$

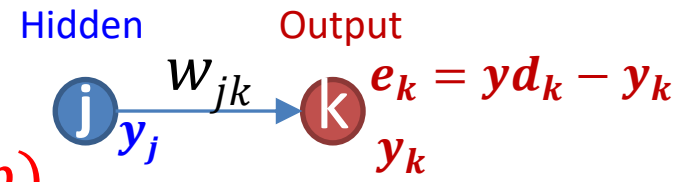
gradiant

diff sigmoid

$$\delta_k(p) = \underline{y_k(p).\left(1 - y_k(p)\right)}.\, e_k(p),$$

$$\Delta w_{jk}(p) = \alpha.\, y_j(p).\, \delta_k(p),$$

$$w_{jk}(p+1) = w_{jk}(p) - \Delta w_{jk}(p)$$



ปรับ weight ตัวที่
เหมือน ex. And gate

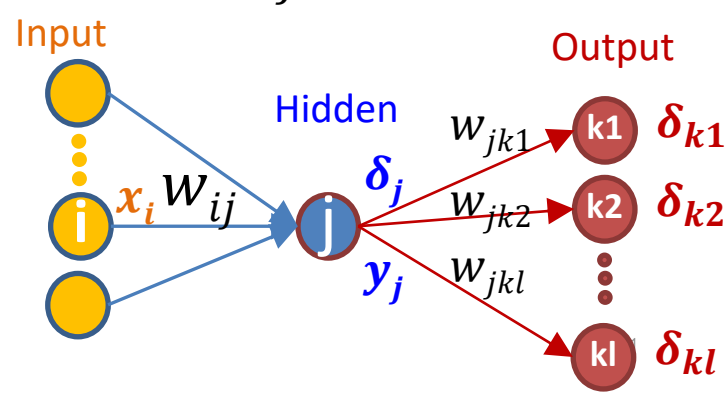
- Calculate the error gradient of neurons in the input-hidden layer.

$$\delta_j(p) = y_j(p).\left(1 - y_j(p)\right).\sum_{k=1}^{l} \delta_k(p).\, w_{jk}(p),$$

learning rate

$$\Delta w_{ij}(p) = \alpha.\, x_i(p).\, \delta_j(p),$$

$$w_{ij}(p+1) = w_{ij}(p) - \Delta w_{ij}(p)$$



Step4: Iterations

- Take the next training pattern, p+1, and go back to step 2. Then repeat the process until the selected error criterion is satisfy.

- A simple stop criterion is when the sum-squared error (SSE) less than a certain number, e.g., 0.1.

sum squared error

$$SSE = \sum_{p=1}^{\#patterns} \sum_{k=1}^{\#outputs} [yd_k(p) - y_k(p)]^2$$

Target actual

$$MSE = \frac{SSE}{\text{จำนวน pattern}}$$

y_j = hidden node
y_k = output node

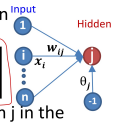
complete training algorithm backpopagation

step 1 : สุ่มset weight กับ thereshold (θ) range (0.5, -0.5)

- Calculate the actual outputs of neurons in hidden layer.

$$y_j(p) = sigmoid\left[\sum_{i=1}^{n}(x_i(p)\cdot w_{ij}(p)) - \theta_j\right]$$

Where n is the number of inputs of neuron j in the



step 2 : calculate output of hidden / output node

hidden node $y_j(p) = sigmoid\left[\sum_{i=1}^{n}(x_i(p)\cdot w_{ij}(p)) - \theta_j\right]$

output node $y_k(p) = sigmoid\left[\sum_{j=1}^{n}(x_j(p)\cdot w_{jk}(p)) - \theta_k\right]$

เอา input x weight แล้วเข้า sigmoid

step 3 : คำนวณ error gradiant ของ hidden / output node เพื่อปรับ weight

ปรับ weight hidden – output layer

$$w_{jk}(p+1) = w_{jk}(p) - \Delta w_{jk}(p)$$

$$\Delta w_{jk}(p) = \alpha \cdot y_j(p) \cdot \delta_k(p)$$

$$\delta_k(p) = \underbrace{\boxed{y_k(p)\cdot(1-y_k(p))}}_{\text{diff sigmoid}} \cdot e_k(p)$$

$$e_k(p) = \underbrace{yd_k(p)}_{\text{target}} - \underbrace{y_k(p)}_{\text{actual}}$$

ปรับ weight input – hidden layer

$$w_{ij}(p+1) = w_{ij}(p) - \Delta w_{ij}(p)$$

$$\Delta w_{ij}(p) = \alpha \cdot x_i(p) \cdot \delta_j(p)$$

$$\delta_j(p) = y_j(p)\cdot(1-y_j(p)) \cdot \sum_{k=1}^{L}(\delta_k(p)\cdot w_{jk}(p))$$

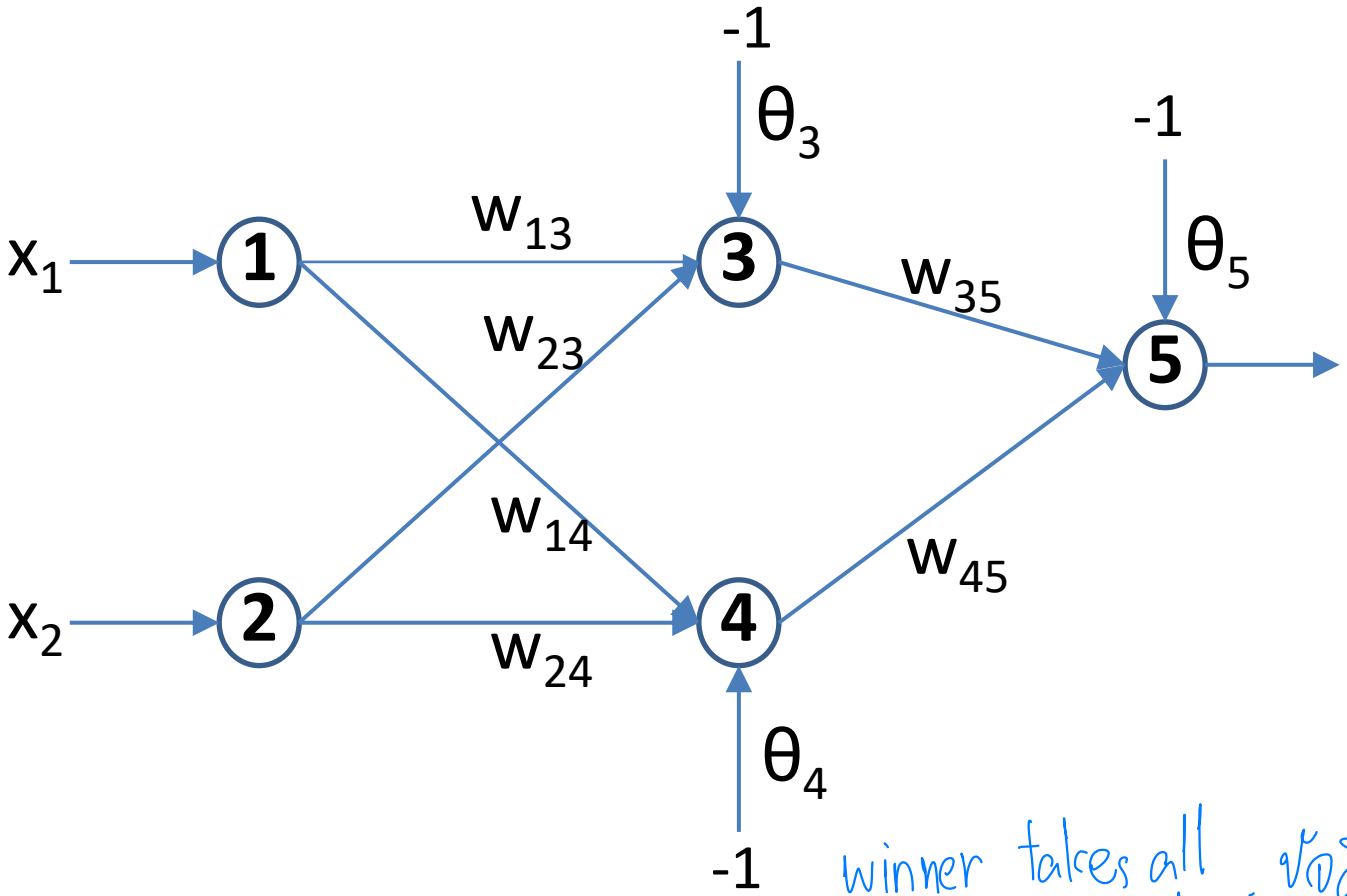
ก็คือ เอา gradiat k มาคูณ weight แต่ละอันให้ครบ แล้วเอามา summation

step 4 : หา error SSE

$$SSE = \sum_{k=1}^{\#pattern}\sum_{k=1}^{\#output}\left[\underbrace{yd_k(p)}_{\text{target}} - \underbrace{y_k(p)}_{\text{actual}}\right]^2$$

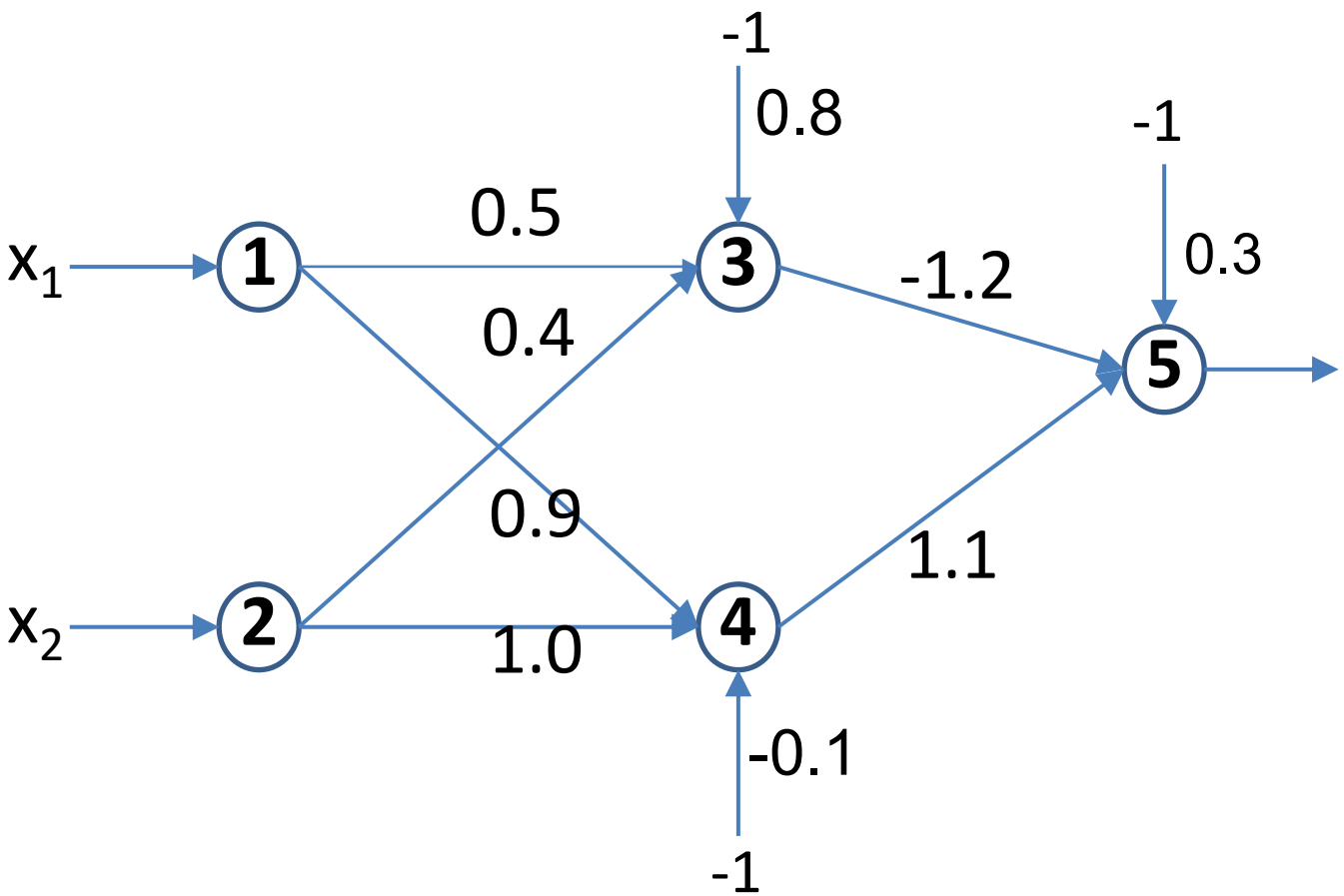
รวม error output node ทุก pattern เข้าด้วยกัน

Example: 3-layer NN for XOR problem



winner takes all ข้อดี
คือมี weight เพิ่มขึ้น (เข้าได้เยอะขึ้น)

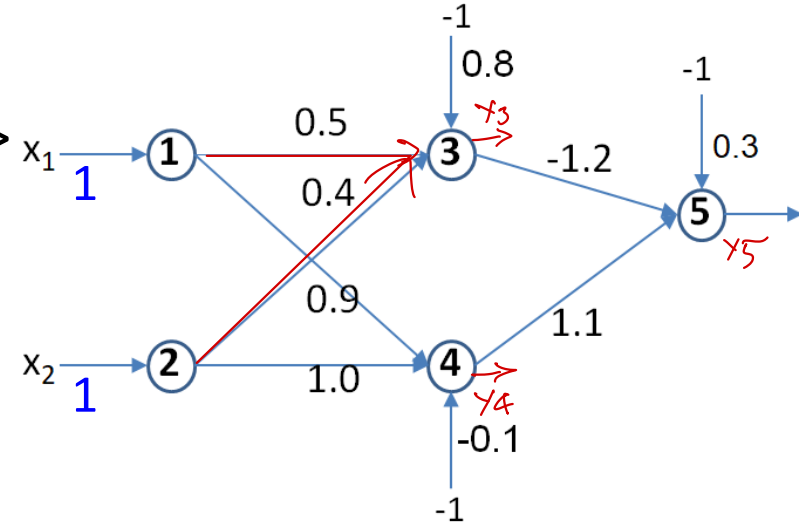
Recall that a single-layer perceptron cannot solve XOR problem.


-1
0.8
-1
0.3
0.5
x_1
1
3
-1.2
0.4
5
0.9
1.1
x_2
2
4
1.0
-0.1
-1

All weights and thresholds are randomly initialized as follows:

Parameter	Initial Value
w_{13}	0.5
w_{14}	0.9
w_{23}	0.4
w_{24}	1.0
w_{35}	-1.2
w_{45}	1.1
θ_3	0.8
θ_4	-0.1
θ_5	0.3

- Consider the training pattern, <1, 1, 0>



$$y_3 = \frac{1}{[1 + e^{-(1*0.5+1*0.4-1*0.8)}]} = 0.5250$$

$$y_4 = \frac{1}{[1 + e^{-(1*0.9+1*1.0-1*0.1)}]} = 0.8808$$

$$y_5 = \frac{1}{[1 + e^{-(-0.525*1.2+0.8808*1.1-1*0.3)}]} = 0.5097$$

$$\mathrm{e} = yd_5 - y_5 = 0 - 0.5097 = -0.5097$$

$$\begin{aligned}\delta_5 &= y_5 \cdot (1 - y_5) \cdot e \\ &= 0.5097 * (1 - 0.5097) * (-0.5097) \\ &= -0.1274\end{aligned}$$

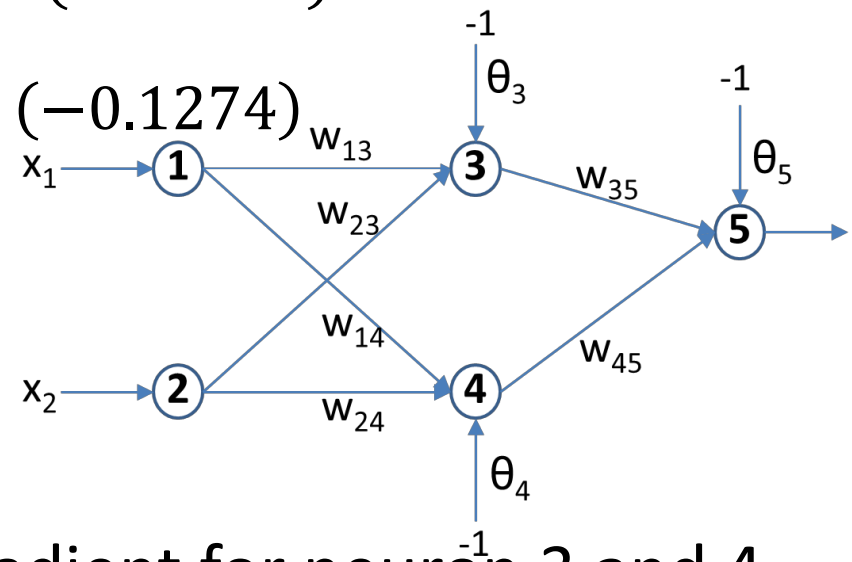
- Assume that the learning rate, α, is equal to 0.1.

$$\Delta w_{35} = \alpha . y_3 . \delta_5 = 0.1 * 0.5250 * (-0.1274) = -0.0067$$

$$\Delta w_{45} = \alpha . y_4 . \delta_5 = 0.1 * 0.8808 * (-0.1274) = -0.0112$$

$$\Delta \theta_5 = \alpha . (-1) . \delta_5 = 0.1 * (-1) * (-0.1274)$$

$$= 0.0127$$



- Next, we calculate the error gradient for neuron 3 and 4.

Diff sigmoid

$$\delta_3 = y_3 . (1 - y_3) . \delta_5 . w_{35}$$

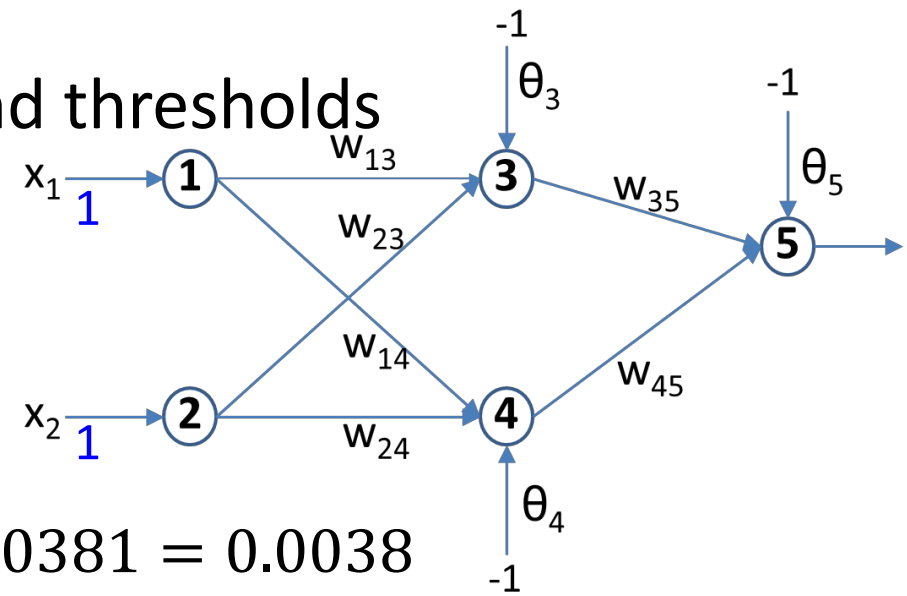
$$= 0.525 * (1 - 0.525) * (-0.1274) * (-1.2) = 0.0381$$

Diff sigmoid

$$\delta_4 = y_4 . (1 - y_4) . \delta_5 . w_{45}$$

$$= 0.8808 * (1 - 0.8808) * (-0.1274) * 1.1 = -0.0147$$

- Calculating Δ of weights and thresholds



$$\Delta w_{13} = \alpha . x_1 . \delta_3 = 0.1 * 1 * 0.0381 = 0.0038$$

$$\Delta w_{23} = \alpha . x_2 . \delta_3 = 0.1 * 1 * 0.0381 = 0.0038$$

$$\Delta \theta_3 = \alpha . (-1) . \delta_3 = 0.1 * (-1) * 0.0381 = -0.0038$$

$$\Delta w_{14} = \alpha . x_1 . \delta_4 = 0.1 * 1 * (-0.0147) = -0.0015$$

$$\Delta w_{24} = \alpha . x_2 . \delta_4 = 0.1 * 1 * (-0.0147) = -0.0015$$

$$\Delta \theta_4 = \alpha . (-1) . \delta_4 = 0.1 * (-1) * (-0.0147) = 0.0015$$

- Lastly, we update all weights and thresholds in the network.

$$w_{13} = w_{13} - \Delta w_{13} = 0.5 - 0.0038 = 0.4962$$

$$w_{14} = w_{14} - \Delta w_{14} = 0.9 + 0.0015 = 0.9015$$

$$w_{23} = w_{23} - \Delta w_{23} = 0.4 - 0.0038 = 0.3962$$

$$w_{24} = w_{24} - \Delta w_{24} = 1.0 + 0.0015 = 1.0015$$

$$w_{35} = w_{35} - \Delta w_{35} = -1.2 + 0.0067 = -1.1933$$

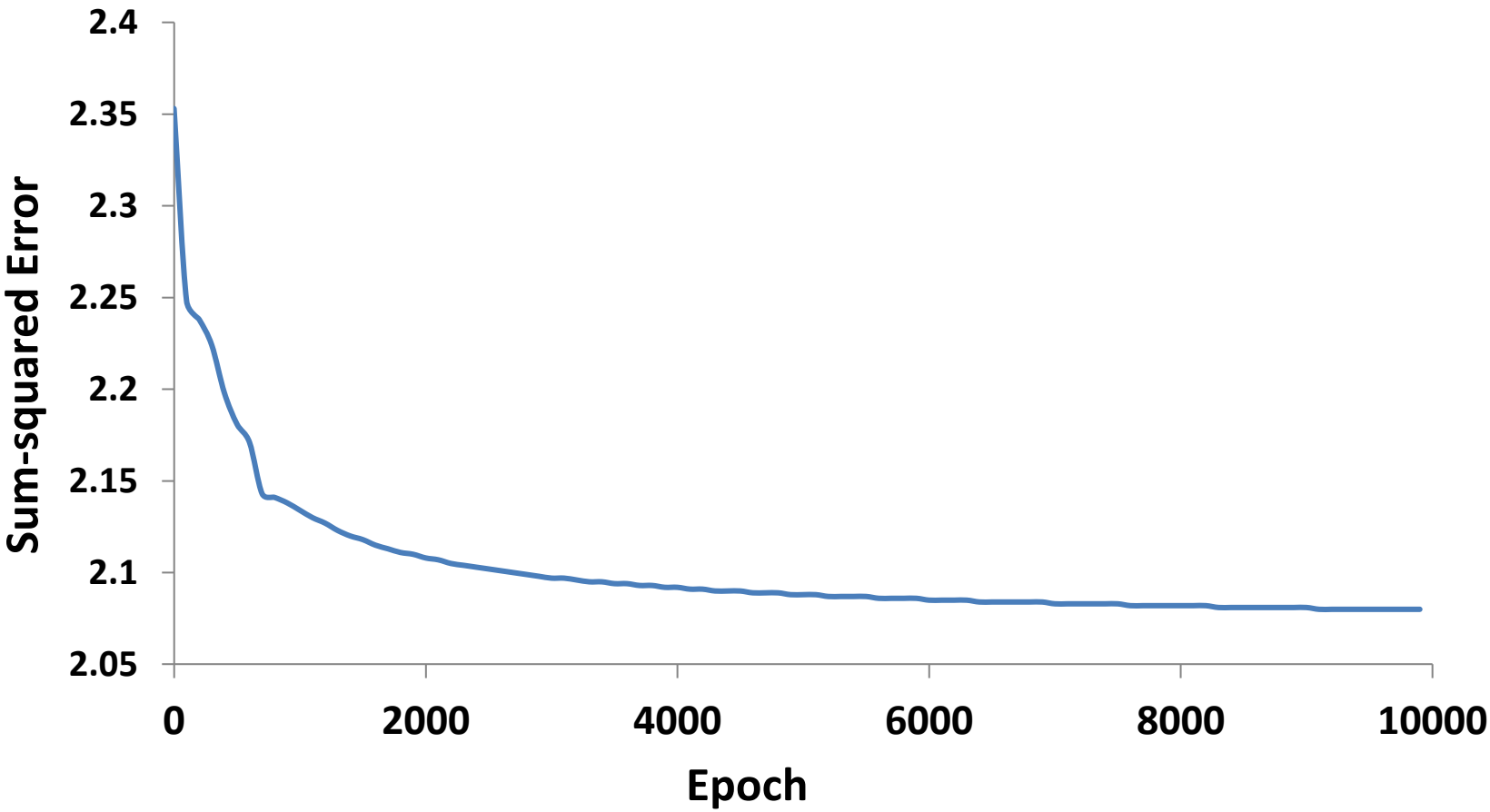
$$w_{45} = w_{45} - \Delta w_{45} = 1.1 + 0.0112 = 1.1112$$

$$\theta_3 = \theta_3 - \Delta\theta_3 = 0.8 + 0.0038 = 0.8038$$

$$\theta_4 = \theta_4 - \Delta\theta_4 = -0.1 - 0.0015 = -0.1015$$

$$\theta_5 = \theta_5 - \Delta\theta_5 = 0.3 - 0.0127 = 0.2873$$

- Repeat the same computation for all training patterns (1 epoch)
- Repeat the process for another epoch until the sum of squared error (SSE) is less than a certain number, e.g. 0.001.



- Sum of squared errors (SSE) of the final network

Input		Desired Output	Actual Output	Error	SSE
x_1	x_2	y_d	y_5	e	
1	1	0	0.0155	-0.0155	0.0010
0	1	1	0.9849	0.0151	
1	0	1	0.9849	0.0151	
0	0	0	0.0175	-0.0175	

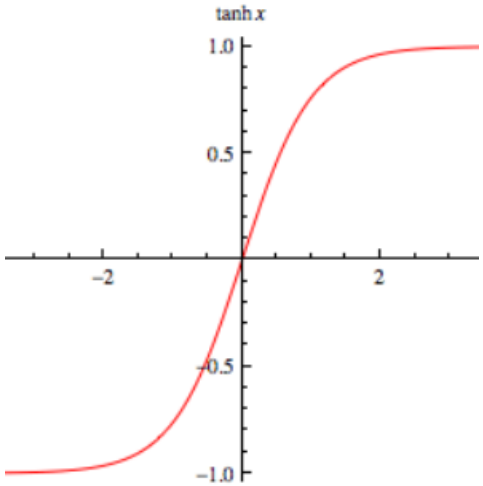
Techniques for improving multilayer NN

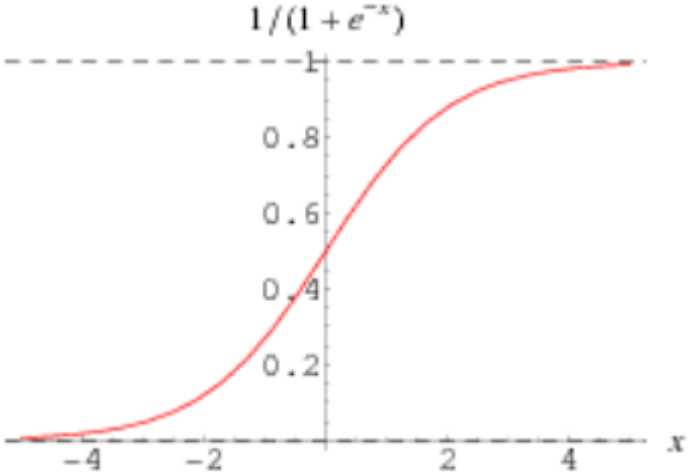
ทำให้ชันขึ้น

- Using a steeper activation function, i.e. tanh (hyperbolic tangent function), to accelerate the convergence.

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

$$= \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} = \frac{e^{2x}-1}{e^{2x}+1}$$





Techniques for improving multilayer NN

- Momentum

The momentum term can reduce the local minima and smooth the variation of the output value.

Momentum

$$\Delta w_{jk} = \boldsymbol{\beta} \Delta w_{jk} - \propto . y_j . \delta_k$$

Where β is a positive number $(0 \leq \beta \leq 1)$ called the momentum constant.

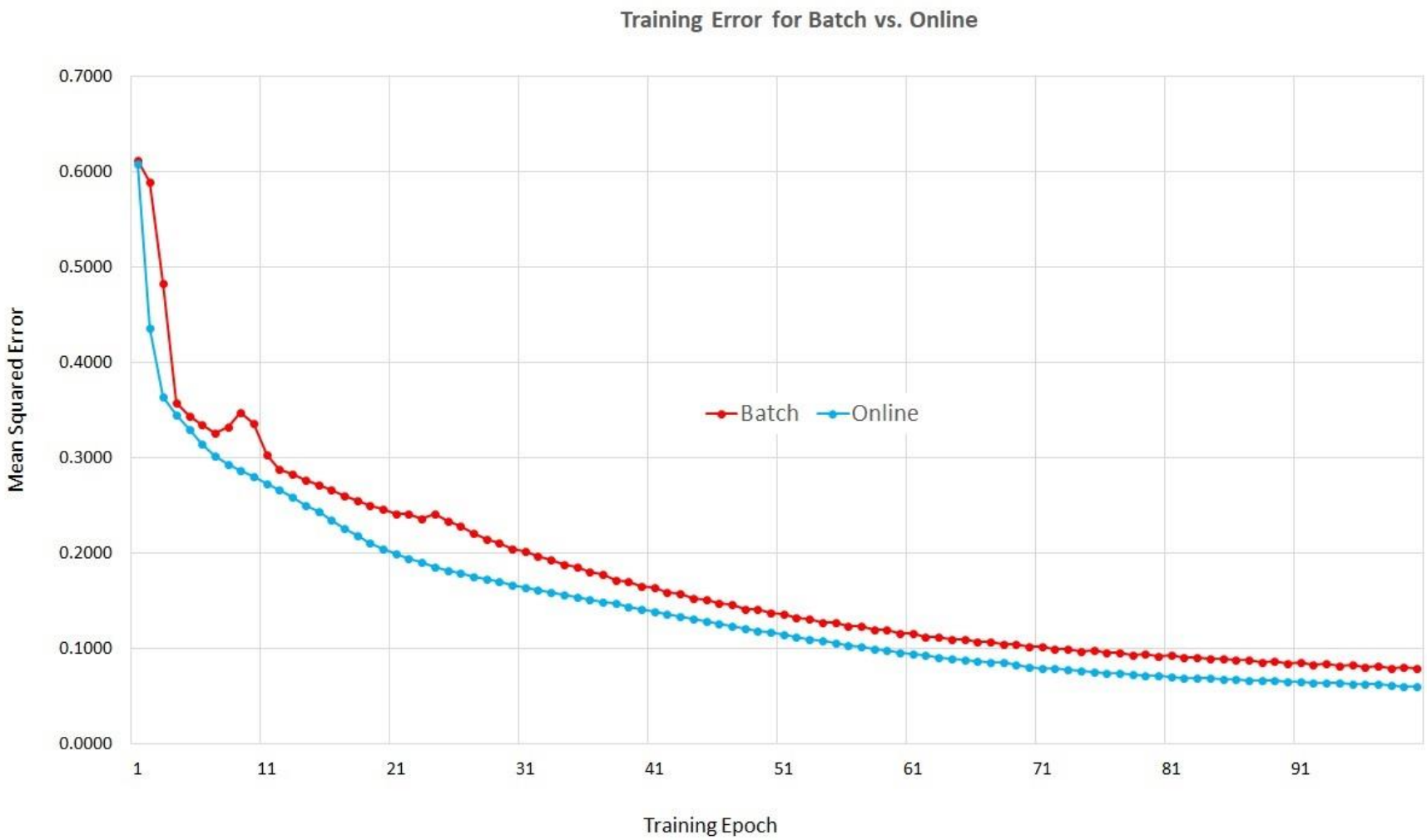
If the momentum term is large then the learning rate should be kept smaller. Otherwise, you might skip the minimum spot with a huge step.

Techniques for improving multilayer NN

- Batch learning train ทีละ 1 pattern แล้วปรับ weight เลย
  - In **online training**, weights and bias values are adjusted for every training item based on the difference between computed outputs and the training data target outputs.
  - In **batch training,** the adjustment delta values are accumulated over all training items, to give an aggregate set of deltas, and then the aggregated deltas are applied to each weight and bias.

    รอบนอก weight ที่เดียว
  - **Online update** = **Batch updates** x No. of samples in training set

    เร็วขึ้น เร็วกว่า
  - The batch algorithm is also slightly more efficient in terms of number of computations.


Training Error for Batch vs. Online
Mean Squared Error
0.7000
0.6000
0.5000
0.4000
0.3000
0.2000
0.1000
0.0000
1
11
21
31
41
51
61
71
81
91
Training Epoch
Batch
Online